Self-Supervised Rotation-Invariant Representations for Unsupervised Motif Discovery in Molecular Dynamics

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Abstract. Identifying local structures in 3D point clouds remains challenging for unsupervised methods, particularly in scientific settings such as molecular dynamics simulations of crystallization, where meaningful motifs correspond to subtle geometric arrangements of atoms. While recent advances in point-cloud representation learning focus primarily on object-level shape understanding, they are less suited to capturing fine-grained local geometric patterns under rotational symmetry. We propose a self-supervised framework for learning rotation-aware representations of atomic neighborhoods and clustering them into local structural motifs. Our method combines a rotation-equivariant encoder with a VICReg objective and a spatially consistent view construction that encourages smooth representations across overlapping neighborhoods. On both synthetic benchmarks and large-scale molecular dynamics simulations, the learned embeddings organize local environments into coherent and interpretable clusters that reveal spatially structured transitions between disordered and ordered atomic configurations. These results demonstrate the potential of self-supervised geometric learning for uncovering physically meaningful structure in large scientific point clouds.

Keywords: Point clouds · Self-supervised learning · Atomic structure

1 Introduction

Understanding local geometric structure in 3D point clouds is central to many problems, including fine-grained part segmentation [18], scene analysis [24], and atomic structure identification [21]. While recent advances have addressed object-level recognition, semantic segmentation, or shape reconstruction, the fully unsupervised discovery of multiple local structural motifs within a single point cloud remains largely unexplored [3,4,12]. This gap is particularly striking in atomic-scale and discrete-particle simulations, where the combinatorial diversity of local

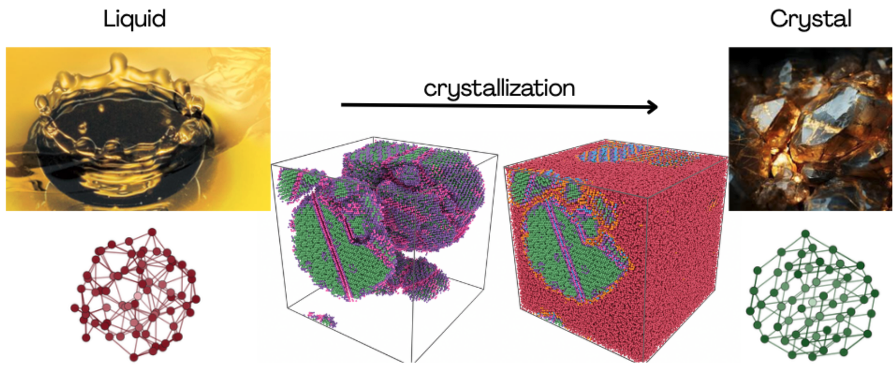


Fig. 1: **Homogeneous crystal nucleation in aluminium: early stages of the phase transition from liquid to crystalline states.** (Left) Typical local atomic structure of the liquid, characterized by disordered atomic arrangements. (Middle) The snapshots illustrate the emergence of nuclei formed by perfect and defective crystalline local structural motifs during the homogeneous nucleation process. The structural identification captures well-defined crystalline nuclei within the liquid (red atoms), including their defects, orientations, and morphologies. (Right) Local structure in a perfect crystalline arrangement. This transition highlights the complexity of the phenomenon, motivating the need for robust unsupervised geometric analysis in molecular dynamics.

atomic structures and the complexity of atomic neighborhoods challenge traditional approaches. Automatically identifying these motifs is crucial for exploring structure-property relationships in systems like high-entropy alloys or metallic glasses, where traditional descriptors fail.

The challenge goes beyond mere identification: in molecular dynamics simulations, classifying the structure of the complete simulation box is not sufficient for capturing phase transformations, *e.g.* crystal nucleation. The objective is rather to cluster local atomic environments into motifs ranging from perfectly ordered, crystal-like configurations to disordered, liquid-like arrangements (see Figure 1), with the latter largely unknown *a priori*. In this way, subtle changes in local structures in transient configurations can be monitored, revealing critical information about the mechanisms that drive phase transformations such as nucleation or glass formation. Capturing these nuances thus demands representations that not only respect geometric symmetries (e.g., SO(3) equivariance) but also vary smoothly with local deformations, a requirement that existing methods struggle to meet in an unsupervised setting.

Learning such representations raises two main challenges. First, atomic neighborhoods are defined up to arbitrary global rotations, so representations must handle SO(3) symmetry appropriately. Standard invariant pipelines, by construction, discard orientation information early in the processing chain, potentially erasing geometrically meaningful information that is critical for distinguishing subtle structural variants. By contrast, equivariant architectures preserve structured features that transform predictably under rotation. Although this simplification may suffice for tasks such as object classification, it is not enough to analyze local atomic environments, where relative orientation often encodes physically relevant information. Enforcing equivariance aligns with the data's symmetry and preserves meaningful directional information during training. Here, a rotation-equivariant Vector Neurons (VN) encoder is used to capture these nuances, deriving invariant embeddings from its equivariant features for clustering to retain structure. Second, the problem is intrinsically local but also highly overlapping and spatially continuous: each atom's neighborhood yields a dense collection of sub-clouds sharing many points. Since crystallization develops spatially, nearby neighborhoods correspond to gradually evolving local structures, and their embeddings should reflect this continuity. Most self-supervised frameworks treat these neighborhoods as independent samples, ignoring their spatial relationships and the gradual evolution of local structures, despite the fact that, in reality, nearby atoms often belong to the same evolving structural motif. This oversight produces embeddings that are either overly rigid (failing to capture smooth evolution) or unstable (collapsing to degenerate solutions). To address this, we adapt a self-supervised strategy from VICReg to an equivariant framework, processing two spatial sub-samplings of each neighborhood through a shared equivariant encoder. Invariant embeddings, derived from these features, are clustered with K-Means without introducing any clustering-specific objective. The result is a latent space where local atomic environments form spatially smooth clusters, faithfully reflecting physical continuity.

The resulting pipeline provides a fully learned, unsupervised alternative to the hand-crafted, descriptor-based pipelines traditionally used for motif discovery in atomistic point clouds [22,14,2,3]. For example, in simulations of aluminium crystallization, our embeddings reveal a smooth transition from liquid-like to FCC-like environments, with intermediate states corresponding to pre-nucleation clusters that are invisible to traditional descriptors (see Section 5). This continuity is critical for understanding mechanisms like homogeneous nucleation or polymorphic phase competition. Our main contributions are: (i) formulating local motif discovery in atomistic point clouds as a fully unsupervised representation learning problem under global SO(3) symmetry with overlapping neighborhoods; (ii) combining a rotation-equivariant VN encoder [7] with a spatially-aware view sampling strategy within the VICReg framework [1], yielding stable invariant embeddings that are both discriminative and spatially smooth without relying on clustering-specific losses; (iii) demonstrating that simple K-Means clustering of the learned embeddings captures ordered, disordered, and intermediate local structures and reveals spatial transition patterns during crystallization; (iv) introducing a synthetic polycrystal benchmark dataset to evaluate local geometric sensitivity in label-scarce settings.

2 Related Work

Our work lies at the intersection of self-supervised learning for 3D point clouds, SO(3)-equivariant representation learning, and unsupervised analysis of local atomic environments.

Self-supervised learning for 3D point clouds. Self-supervised learning (SSL) has become a standard paradigm for learning representations without manual annotations. In computer vision, contrastive methods such as SimCLR [6] learn invariances by maximizing agreement between augmented views, while non-contrastive objectives such as Barlow Twins [30] and VICReg [1] combine view consistency with redundancy reduction and variance preservation. For 3D point clouds, PointContrast [28] applies contrastive pre-training to large-scale 3D scenes, while Point-BERT [29], Point-MAE [16], PointM2AE [32], and PointGPT [5] adapt masked modeling or autoregressive pre-training to point sets. More recently, rotation-robust variants such as RI-MAE [23] incorporate rotational invariance into self-supervised point cloud learning. These approaches primarily focus on object-level representation learning for supervised tasks. In contrast, our goal is to directly organize local neighborhoods into clusterable embeddings.

SO(3)-equivariant architectures for geometric learning. Rotational symmetry is central to 3D learning, inspiring extensive work on equivariant neural networks. Tensor Field Networks [25], SE(3)-Transformers [9], and related architectures enforce consistent feature transformation under rigid motions. Vector Neurons (VN) [7] offer a lightweight, practical alternative for point clouds by using 3D vector features. Equivariance has also been paired with self-supervision, as in ConDor [20], which learns canonical poses for shapes. While effective when a stable reference frame exists, these methods can struggle with highly symmetric or locally isotropic neighborhoods, common in atomistic data.

Local representation learning in atomistic systems. In atomistic simulations, local structure has traditionally been characterized using hand-crafted descriptors such as Steinhardt bond-order parameters [22], averaged bond-order parameters [14], and SOAP-like representations [2]. While highly successful for materials analysis, these descriptors require a priori choices about which geometric features to encode. However, more recent work explores the unsupervised discovery of local atomic environments directly from simulation data. Existing approaches combine structural descriptors with clustering or manifold learning techniques to identify representative local environments and defects [31,4,3,19,12]. Other work investigates large-scale pre-training of geometric graph networks to learn transferable atomistic representations, primarily for transfer or zero-shot analysis rather than clustering of local motifs [17].

Position of our work. These works highlight the promise of unsupervised atomistic representation learning, but most existing pipelines still rely on fixed descriptors, either hand-crafted structural descriptors, topological summaries, or

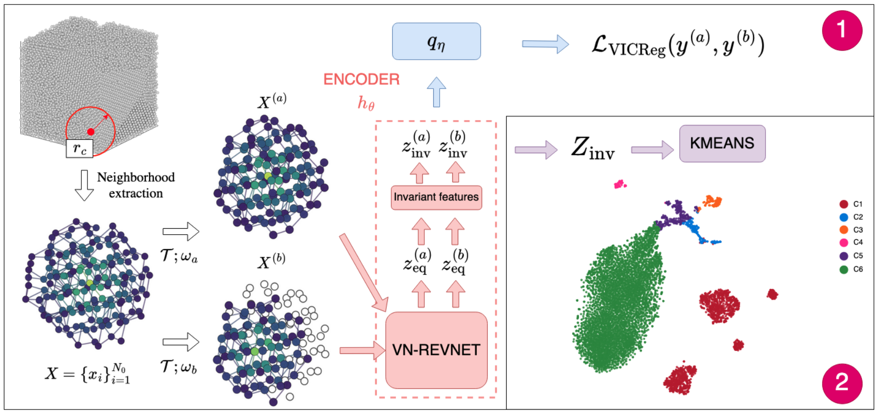


Fig. 2: **Overview of the self-supervised learning pipeline for motif discovery in molecular dynamics.** (1) Self-supervised training: A neighborhood X of N_0 atoms is extracted from the simulation box and augmented into two views, $X^{(a)}$ and $X^{(b)}$, using transformations $\mathcal{T}$. The VN-RevNet encoder processes these views to produce equivariant z_{eq} and invariant z_{inv} embeddings. The VICReg loss $\mathcal{L}_{\mathrm{VICReg}}$ is applied to the projected embeddings $y^{(a)}$ and $y^{(b)}$ to learn robust representations. (2) Clustering: The invariant embeddings Z_{inv} are clustered using K-Means, revealing distinct structural motifs in the data. The resulting clusters are visualized using t-SNE, demonstrating the method's ability to capture local geometric patterns without supervision.

features inherited from atomistic potentials, followed by clustering or manifold analysis. In contrast, our approach learns the representation directly from raw local coordinates using self-supervision while explicitly respecting global SO(3) symmetry through an equivariant encoder. Furthermore, unlike most prior SSL methods for point clouds, our setting involves dense collections of strongly overlapping neighborhoods extracted from the same atomic configuration. Nearby samples should therefore remain geometrically and spatially consistent in the latent space. This combination of equivariant self-supervision, overlapping local neighborhoods, and clustering-oriented representation learning defines the regime we target in this work.

3 Proposed Method

We present a self-supervised framework for learning rotation-invariant representations of atomic neighborhoods. An equivariant encoder produces invariant embeddings trained with VICReg on augmented views, after which clustering reveals local motifs. Figure 2 illustrates the overall pipeline.

3.1 Problem Statement and Neighborhood Construction

Setup. Given an atomic configuration represented as a point cloud, we aim to discover recurring local geometric motifs without supervision. Rather than operating on entire configurations, our approach focuses on local atomic neighborhoods, which capture the structural environments around individual atoms.

Formally, let $X = \{x_i \in \mathbb{R}^3\}_{i=1}^{N_0}$ denote a local neighborhood extracted from a configuration. Our goal is to learn two complementary representations: (i) a rotation-equivariant latent z_{eq} that transforms predictably under rotations, and (ii) a rotation-invariant embedding z_{inv} suitable for clustering.

We therefore learn an encoder h_θ, parameterized by θ

$$(z_{\mathrm{inv}}, z_{\mathrm{eq}}) = h_\theta(X), \quad \text{s.t.} \quad z_{\mathrm{eq}}(RX) = R\, z_{\mathrm{eq}}(X), \quad z_{\mathrm{inv}}(RX) = z_{\mathrm{inv}}(X)$$

for any rotation $R \in \mathrm{SO}(3)$. Enforcing equivariance in z_{eq} acts as a geometric inductive bias that preserves directional information during representation learning [26], rather than collapsing it prematurely into rotation-invariant scalars. The invariant embedding z_{inv} is subsequently derived from z_{eq} through invariant operations described in Section 3.2.

Atomic neighborhood construction. For each atom, we collect all atoms within a cutoff radius r_c. If more than N_0 atoms fall within this radius, we retain the N_0 closest ones. This defines a point cloud $X = \{x_i \in \mathbb{R}^3\}_{i=1}^{N_0}$. Each neighborhood is centered on its central atom, and coordinates are normalized by r_c, keeping magnitudes typically below one. This procedure yields a dense set of overlapping neighborhoods across the configuration.

3.2 Equivariant Encoder and Invariant Embeddings

To capture the geometric structure of atomic neighborhoods while respecting rotational symmetry, we employ a rotation-equivariant neural encoder based on Vector Neurons (VN) [7]. VN networks represent intermediate features as 3D vectors and ensure that these features transform consistently under rotations of the input. Given a neighborhood X, the encoder produces a set of equivariant vector features $z_{\mathrm{eq}} = f_\theta(X)$, where each feature channel corresponds to a vector in $\mathbb{R}^3$. Depending on the backbone configuration, these features can be either global, $z_{\mathrm{eq}} \in \mathbb{R}^{B \times C \times 3}$, or per-point, $z_{\mathrm{eq}} \in \mathbb{R}^{B \times N_{\mathrm{model}} \times C \times 3}$, where B denotes the batch size, C the number of feature channels, and N_{model} the number of points provided to the encoder for each neighborhood.

Invariant feature construction. To obtain rotation-invariant descriptors suitable for clustering, we derive an invariant embedding from the equivariant features. Let $z_{\mathrm{eq}} = (v_1, \ldots, v_C)$ with $v_i \in \mathbb{R}^3$ denote the vector channels produced by the equivariant encoder. We construct scalar invariants from these vectors using rotation-invariant operations. First, we compute channel-wise norms

$n(v_i) = \|v_i\|_2$, which are unchanged under global rotations. Optionally, additional geometric information can be captured through pairwise dot products $d(v_i, v_j) = \langle v_i, v_j \rangle$. The resulting scalars are concatenated into a feature vector

$$g = [\, n(v_1), \ldots, n(v_C),\, d(v_{i_1}, v_{j_1}), \ldots],$$

which is then normalized to produce the invariant embedding $z_{\text{inv}} = \text{Norm}(g)$.

3.3 Self-Supervised Training

The encoder is trained in a self-supervised manner using the VICReg objective [1]. Given an input neighborhood X, we generate two views

$$X^{(a)} = \mathcal{T}(X; \omega_a), \qquad X^{(b)} = \mathcal{T}(X; \omega_b),$$

where $\mathcal{T}$ denotes a stochastic view generator and ω_a, ω_b are independent random draws. The two views are processed by a shared encoder h_θ to obtain invariant embeddings, i.e., $z_{\text{inv}}^{(a)} = h_\theta(X^{(a)})$ and $z_{\text{inv}}^{(b)} = h_\theta(X^{(b)})$. During training, the embeddings are further mapped by a projection head q_η, such that $y^{(a)} = q_\eta(z_{\text{inv}}^{(a)})$ and $y^{(b)} = q_\eta(z_{\text{inv}}^{(b)})$.

The VICReg objective combines three complementary terms that enforce view consistency, prevent dimensional collapse, and reduce redundancy across feature dimensions:

$$\mathcal{L}_{\text{VICReg}} = \lambda_{\text{inv}}\mathcal{L}_{\text{inv}} + \lambda_{\text{var}}\mathcal{L}_{\text{var}} + \lambda_{\text{cov}}\mathcal{L}_{\text{cov}}.$$

$\mathcal{L}_{\text{inv}}$ encourages agreement between embeddings of the two views, while $\mathcal{L}_{\text{var}}$ and $\mathcal{L}_{\text{cov}}$ ensure sufficient feature variance across the batch and reduce correlations between embedding dimensions.

View construction. The view operator $\mathcal{T}$ generates views of a neighborhood through spatial resampling. Starting from a neighborhood centered on an atom c, we randomly choose to keep the original center or to shift the center to one of its neighboring atoms. Let c' denote the selected center. The new neighborhood is then defined by selecting the N_{model} atoms closest to c':

$$X' = \text{TopK}_{N_{\text{model}}}\left(\|x_i - c'\|\right), \quad x_i \in X.$$

In practice, we first sample N_{sample} points from the N_0 points of the neighborhood and then retain the N_{model} closest points after the potential center shift. For the main experiments, we use $N_{\text{sample}} = 128$ and $N_{\text{model}} = 80$. This neighbor-centered resampling produces pairs of views corresponding to slightly shifted local environments. Since neighboring atoms share many nearby points, the resulting views strongly overlap while exhibiting small geometric variations. As a consequence, nearby neighborhoods tend to produce nearby embeddings, promoting spatial smoothness in the learned representation.

In addition to center-shift resampling, we optionally apply small geometric perturbations (e.g., coordinate jitter or random point removal) that increase view diversity while preserving the identity of the local atomic motif.

3.4 Clustering Protocol

Clustering has been widely studied and many approaches have been developed for a variety of circumstances. We focus on the standard clustering algorithm K-Means, but other clustering algorithms can be used. For a fixed number of clusters k, the k-means algorithm takes a set of vectors as input, in our case the normalized invariant embeddings z_{inv}, and clusters them into k distinct groups based on a geometric criterion. This jointly learns a centroid matrix C and the cluster assignments $(c_n)_{n=1,\dots,N}$ of each local structure:

$$\min_{C} \frac{1}{N} \sum_{n=1}^{N} \min_{c_n \in \{0,1\}^k} \|z_{\mathrm{inv}}^{(n)} - C c_n\|_2^2 \text{ such that } c_n^T \mathbf{1}_k = 1.$$

When labels are available, we report ACC, NMI, and ARI on the resulting assignments; when no ground truth is available, we interpret the clusters as homogeneous groups.

4 Experiments on the Synthetic Polycrystal Benchmark

We compare our method to state-of-the-art methods on a novel synthetic polycrystal benchmark. The code and configuration files required to reproduce the benchmark dataset are included in the paper's linked code repository.

4.1 Synthetic polycrystal benchmark

Quantitative evaluation of motif discovery in molecular dynamics (MD) is hindered by the lack of ground-truth labels and the need to capture rotation-invariant local geometric cues, a task for which standard self-supervised learning (SSL) methods are ill-suited. Unlike object-level benchmarks, where classes differ by coarse topology, MD data consists of metrically similar point clouds distinguished solely by angular correlations in neighbor shells. To address this gap, we introduce a controlled polycrystal benchmark that isolates three key challenges:

(i) *Known ground-truth motifs*: a cubic simulation box is tessellated into $G = 16$ Voronoi grains, each randomly assigned one of four phases (three ordered, BCC/FCC/HCP, and one disordered, amorphous). All phases share a nearest-neighbor distance of 2.49 Å and iron-like densities.

(ii) *Decoupling motif identity from orientation*: crystalline grains are created by tiling unit cells with random rotations $R \sim \mathrm{SO}(3)$, while the amorphous phase is refined to match realistic short-range order (coordination ≈ 11).

(iii) *Realistic disorder*: to replicate ambiguous neighborhoods near grain boundaries, we introduce thermal displacements (Debye model at $T = 325$ K, calibrated to iron); localized rotation bubbles (rigid micro-rotations of $\approx 12°$ in spheres of radius 7 Å); point vacancies (0.8% dropout + relaxation) and density bubbles (local atom depletion/concentration).

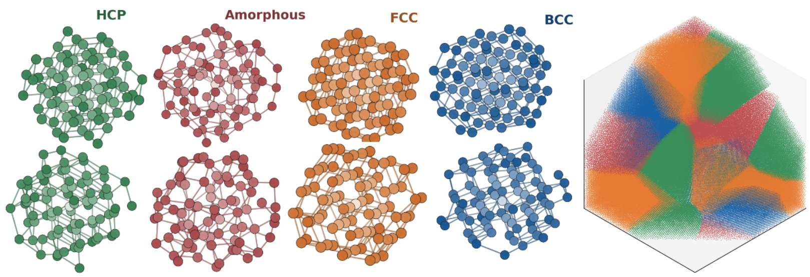


Fig. 3: **Synthetic polycrystal benchmark**. (Left) Representative 80-point local environments for the four classes: FCC (face-centered cubic), BCC (body-centered cubic), HCP (hexagonal close-packed), and amorphous, shown in distinct colors. For each class, the upper example is the pure structure and the lower example is a deformed version. (Right) Synthetic polycrystal simulation box colored by class (diagonal cut through the full box).

We sample local neighborhoods as point clouds using a grid with spacing $\Delta = 2r_{\text{cut}}(1 - K_{\text{overlap}})$, centering each neighborhood on an atom and normalizing coordinates by r_{cut}. After excluding centers near the box boundary to avoid edge artifacts, we retain the $N_0 = 128$ nearest atoms per center. Neighborhoods are further filtered to include only those with atoms from a single grain, yielding a clean 4-class benchmark (BCC/FCC/HCP/amorphous) focused on motif identity.

With 50,653 neighborhoods, this benchmark reveals the limitations of existing SSL methods that rely on global pose cues, and it establishes a rigorous framework to evaluate local geometric sensitivity in label-scarce MD. It is illustrated in Figure 3.

4.2 Implementation details

We use a VN-RevNet encoder [15] to extract equivariant features from local neighborhoods. Invariant embeddings are obtained by taking the norms of the equivariant vector channels, producing representations of dimension $L = 240$. During self-supervised training, each sample generates two augmented views, including volume-preserving strain, occlusion, coordinate jitter, and neighbor-centered re-centering on one of the $k = 6$ nearest atoms. Each view contains $N_{\text{model}} = 80$ points. The model is trained with VICReg [1]. At inference time, the projector is discarded and clustering is performed on frozen invariant embeddings z_{inv} using K-Means with `K-Means++` initialization.

We compare against classical descriptors used as fixed representations and clustered by K-Means, namely Steinhardt, CNA, and SOAP. We also compare against self-supervised point-cloud baselines including PointGPT [5], Point-M2AE [32], PointContrast [28], and RI-MAE [23]. All learned baselines are reimplemented in a shared PyTorch Lightning training and evaluation pipeline and

	Canonical			SO(3)-randomized		
Method	ACC	NMI	ARI	ACC	NMI	ARI
Steinhardt (Q_ℓ)	0.89±.005	0.69±.009	0.73±.011	0.89±.005	0.69±.009	0.73±.010
CNA	0.68±.006	0.50±.008	0.43±.008	0.68±.006	0.50±.008	0.43±.008
SOAP	0.92±.004	0.77±.009	0.79±.007	0.92±.004	0.77±.009	0.79±.007
PointGPT [5]	0.35±.04	0.10±.04	0.09±.03	0.28±.04	0.08±.03	0.05±.02
Point-M2AE [32]	0.46±.03	0.25±.04	0.18±.03	0.47±.04	0.23±.04	0.17±.03
PointContrast [28]	0.85±.02	0.71±.06	0.69±.06	0.85±.02	0.71±.06	0.70±.06
RI-MAE [23]	0.80±.03	0.58±.05	0.34±.05	0.80±.03	0.58±.05	0.34±.05
Ours	**0.95±.03**	**0.84±.08**	**0.88±.08**	**0.95±.04**	**0.84±.09**	**0.88±.08**

Table 1: **Unsupervised classification on the synthetic polycrystal benchmark.** Clustering quality of classical physically motivated baselines, self-supervised point-cloud baselines, and our rotation-invariant embeddings. Clustering is performed with K-Means ($K = 4$) and Hungarian matching for ACC. Reported results correspond to means ±std computed on 5 runs. Canonical: default dataset orientation. SO(3)-randomized: each test shape is rotated by a random $R\sim\mathrm{SO}(3)$.

trained on the balanced synthetic polycrystalline benchmark using the same data split, normalization, model-selection criterion, and SO(3) robustness protocol. We retain the best run for each method after a targeted tuning pass. Because self-supervised pretraining losses are not directly comparable across methods, we report only clustering-quality metrics. Full baseline architecture, training, and tuning details are provided in the supplementary material.

Code availability. The code for our method and experiments is available at `github.com/maloq/md-motif-ssl`.

4.3 Results on the synthetic polycrystal benchmark

Table 1 reports clustering results on the synthetic polycrystal benchmark. Our method outperforms all baselines by a clear margin across all metrics (ACC/NMI/ARI), indicating that the representation learned by our approach is meaningful for the clustering task. While only RI-MAE is explicitly rotation-invariant, several other methods perform similarly in the canonical and SO(3)-randomized settings because training-time rotation augmentation already provides some robustness. However, an equivariant design is preferable in our context.

4.4 Ablation study on the synthetic polycrystal benchmark

We now present the ablation study over the encoder, the proposed views, the number of sampled points, the invariant embedding dimension, and the self-supervised loss.

Table 2: **Encoder ablation on the synthetic polycrystal dataset.** All encoders are trained with the same self-supervised protocol and evaluated by clustering frozen invariant embeddings with $K = 4$. Reported results correspond to means ±std computed on 5 runs.

Encoder	Canonical			SO(3)-randomized		
	ACC	NMI	ARI	ACC	NMI	ARI
PointNet	0.51±0.07	0.29±0.07	0.23±0.09	0.43±0.02	0.22±0.02	0.17±0.02
VN-PointNet	0.78±0.10	0.73±0.07	0.73±0.09	0.77±0.10	0.70±0.10	0.69±0.13
DGCNN	0.86±0.06	0.70±0.08	0.70±0.16	0.62±0.06	0.55±0.05	0.49±0.09
VN-DGCNN	0.93±0.02	0.79±0.04	0.81±0.05	0.92±0.02	0.79±0.04	0.81±0.04
VN-RevNet	**0.95±0.03**	**0.84±0.08**	**0.88±0.08**	**0.95±0.04**	**0.84±0.08**	**0.88±0.08**

Encoder. For the encoder, we compare VN-RevNet with competitive architectures, including PointNet [18] and DGCNN [27], along with their VN variants (VN-PointNet and VN-DGCNN). The results are summarized in Table 2. As expected, PointNet and DGCNN are rotation-variant when operating directly on coordinates, leading to a clear performance drop under SO(3)-randomized evaluation. DGCNN remains competitive in the canonical setting thanks to EdgeConv capturing local geometric patterns, but without explicit rotation constraints, it does not generalize across arbitrary global rotations. Overall, VN architectures provide the most robust representations across both canonical and randomized poses, with VN-RevNet being the most effective.

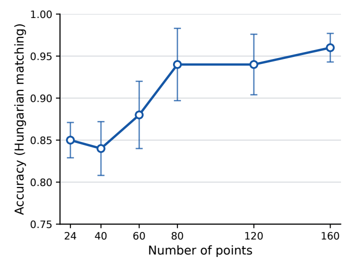
Views. We study the role of view construction with a leave-one-out ablation (Fig. 4a). Removing the neighbor view causes the largest performance drop, highlighting its importance for learning consistent representations across overlapping neighborhoods. In contrast, removing strain or jitter has a more moderate effect.

Evolution with respect to the number of points. We study how performance varies with the number of model input points N_{model}, ranging from 24 to 160. To preserve the neighborhood structure, we keep the ratio N_{model}/N_0 fixed at 0.625, scaling N_0 accordingly (Figure 4b). Very small values of N_{model} provide insufficient geometric information, while performance improves steadily beyond $N_{\text{model}} = 80$. We therefore select $N_{\text{model}} = 80$ as a trade-off between performance and spatial resolution.

The invariant dimension controls the number of scalar features extracted from the equivariant representation. Performance improves substantially from $L = 64$ to $L = 240$, after which it saturates or slightly decreases, suggesting that $L = 240$ offers a favorable trade-off between expressivity and redundancy. We also compare three self-supervised redundancy-reduction objectives. W-MSE performs substantially worse in our setting. This suggests that batch whitening may be too restrictive for dense collections of overlapping atomic neighborhoods, where correlated feature dimensions can still be useful for capturing local geometric motifs. Barlow Twins and VICReg produce more clusterable representa-

Experiment	ACC	NMI	ARI
Baseline (all active)	0.95±0.03	0.84±0.08	0.88±0.08
Neighbor both views	0.94±0.03	0.84±0.08	0.86±0.04
No neighbor view	0.90±0.04	0.82±0.06	0.81±0.06
No jitter	0.92±0.02	0.83±0.05	0.84±0.04
No strain	0.93±0.02	0.83±0.05	0.84±0.05

(a)



(b)

Fig. 4: **Ablation of view construction and neighborhood size.** (a) Leave-one-out ablation of the VICReg view augmentations (mean ±std computed on 5 runs). (b) Clustering accuracy as a function of the number of sampled points N_{model} used in the views.

tions, with VICReg giving the best overall performance and being less sensitive to hyperparameter tuning in our experiments.

5 Real Molecular Dynamics Experiments

Experimental setup. Our method is applied to the homogeneous nucleation pathway of aluminium studied in a recent work [3], generated by large-scale molecular dynamics simulations with LAMMPS using an embedded atom model (EAM) potential. The simulation contains 10^6 atoms in a fully periodic box. A liquid configuration above the melting point is rapidly quenched below the melting point in the undercooled regime, and homogeneous nucleation is then monitored at $T = 650$ K near the nose of the time-temperature-transformation curve [3,11]. The selected simulation output consists solely of atomic coordinates at six time points: 166, 170, 174, 175, 177, and 240 ps, spanning the transition from a supercooled liquid to a fully crystallized polycrystal. Additional details are provided in the supplementary material.

Local neighborhoods are extracted with cutoff radius $r_{\text{cut}} = 7.2$ Å using grid-based sampling with overlap parameter $K_{\text{overlap}} = 0.1$. Central atoms are chosen near the nodes of a cubic grid with spacing $\Delta = 2r_{\text{cut}}(1 - K_{\text{overlap}})$. For each selected center, atoms within r_{cut} are collected and truncated to $N_0 = 128$ points; during training, each view further subsamples $N_{\text{model}} = 80$ points. Across the six snapshots, this procedure yields 1,738,803 local neighborhoods. Thus, the real-MD experiment evaluates the method in a large-scale unlabeled regime where local structures are sampled densely across liquid, interfacial, defective, and crystalline regions.

Latent space and unsupervised clustering. Clustering of the real MD neighborhoods is performed with K-Means using $K = 6$ to obtain a sufficient diversity of local structures involved in the formation of crystalline nuclei. The

Variant	ACC	NMI	ARI
Invariant embedding dimension L			
64	0.81	0.55	0.57
160	0.91	0.80	0.76
240	**0.95**	**0.84**	**0.88**
320	0.94	0.83	0.84
728	0.93	0.82	0.84
Self-supervised objective			
W-MSE [8]	0.71	0.40	0.39
Barlow Twins [30]	0.90	0.80	0.81
VICReg [1]	**0.95**	**0.84**	**0.88**

Table 3: **Ablation of invariant embedding dimension and self-supervised objective.** All variants are evaluated on the synthetic polycrystal benchmark by K-Means clustering of frozen invariant embeddings with $K = 4$.

K-Means model is fitted on local structures sampled from all six simulation boxes mentioned above, reflecting the expectation that the trajectory contains not only liquid-like and crystal-like environments but also intermediate structures associated with nucleation fronts and defects. Importantly, clustering is performed purely in the learned embedding space, without relying on handcrafted structural descriptors such as bond-order parameters [22], common neighbor analysis (CNA) [10], or polyhedral template matching (PTM) [13].

Figure 5 presents a t-SNE projection of the invariant embeddings, with representative neighborhoods near cluster centers. The latent space clearly separates disordered, ordered, and intermediate motifs into distinct groups. While we do not enforce direct alignment with the clusters in [3], we use it as a physical reference. That study shows that aluminium crystallization follows a one-step pathway: embryos form in low-five-fold-symmetry liquid regions, growing with FCC ordering and often featuring patchy morphology and stacking faults.

Interpretation of the discovered clusters. Cluster C_6 corresponds to the most disordered, liquid-like environment and dominates the early snapshots. At the opposite end, cluster C_1 contains the most ordered crystalline neighborhoods and becomes the dominant motif in the final state. The remaining clusters C_2–C_5 occupy intermediate positions both in latent space and in the simulation box and can be interpreted as partially ordered, defective, or interfacial structures.

Among these intermediate motifs, C_5 appears early during nucleation and forms extended shells around the growing crystalline regions before remaining in the final polycrystal. This behavior is consistent with a near-crystalline interfacial motif, i.e. a distorted close-packed environment surrounding the ordered crystal core during growth and later persisting at grain boundaries. The weakly populated clusters C_2–C_4 are located on crystal surfaces, internal defective re-

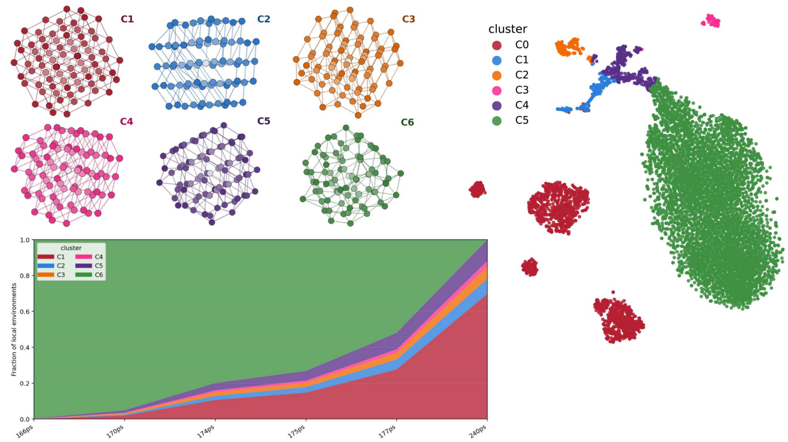


Fig. 5: **Latent-space organization and temporal evolution of the discovered clusters during crystallization on the real MD dataset.** *Top left*: representative neighborhoods for the six discovered clusters, selected as samples nearest to the cluster centers in the learned embedding space. *Right*: t-SNE projection of the invariant embeddings z_{inv}, colored by K-Means cluster assignment. *Bottom left*: fraction of neighborhoods assigned to each cluster over time.

gions, and boundary-like structures; following the reference analysis of [4], they are plausibly associated with defective close-packed motifs, including stacking-fault-rich or otherwise distorted crystalline environments. We keep this interpretation deliberately cautious, since the clusters are discovered without supervision and are not constrained to match a predefined structural taxonomy.

This interpretation is supported by the temporal evolution of the cluster populations shown in Figure 5. At 166 ps, the system is almost entirely assigned to C_6, confirming that the initial configuration is dominated by liquid-like neighborhoods. As crystallization proceeds, the fraction of C_6 decreases monotonically while the population of C_1 grows steadily. By the final frame at 240 ps, C_1 accounts for roughly 70% of the neighborhoods, whereas C_6 has nearly disappeared. The intermediate clusters remain individually subdominant but together represent a substantial fraction of the system during nucleation and growth, and persist in the final polycrystal. This behavior is consistent with clusters encoding interfaces, stacking-faulted regions, and grain-boundary environments rather than bulk liquid or bulk crystal.

Spatial evolution during crystallization. The spatial distribution of the clusters is shown in Figure 6. Early frames are dominated by the liquid-like background (C_6). As crystallization begins, compact regions of ordered and partially ordered environments emerge within this background, corresponding to crystalline nuclei and their surrounding transition layers. These regions grow, impinge, and progressively fill the simulation box. The second row removes the

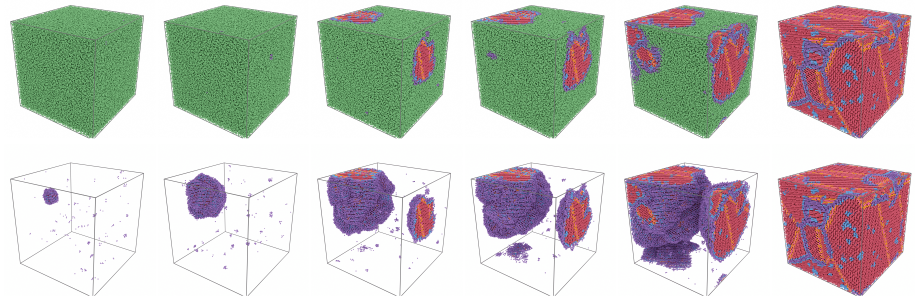


Fig. 6: **Spatial evolution of the identified clusters during homogeneous nucleation.** Columns correspond to successive simulation times from 166 ps to 240 ps. *Top row*: full simulation box colored by cluster assignment. The early frames are dominated by the liquid-like cluster C_6, while ordered and intermediate clusters progressively nucleate and expand. *Bottom row*: the same snapshots with the liquid-like cluster removed, highlighting the emergence, growth, and impingement of ordered and interfacial motifs. The final configuration is a polycrystal composed of large ordered domains enriched in C_1, separated by boundary regions where intermediate clusters, especially C_5, remain prominent.

dominant liquid-like background and highlights only the ordered and intermediate clusters, which makes the nucleation-and-growth process clearer. Early frames show compact precrystalline islands dominated by C_5, while a crystalline core (C_1) develops inside these regions as growth proceeds. At later times, ordered domains expand and eventually impinge, producing a final polycrystal in which C_1 occupies grain interiors and the intermediate clusters are enriched at liquid–nucleus interfaces during growth and at grain boundaries in the final polycrystal (see the interpretation of clusters above). This spatial organization is consistent with the physical picture reported for the same aluminium trajectory in [3], where nucleation proceeds through direct FCC ordering followed by growth of patchy nuclei containing stacking faults and interfacial distortions. Notably, this interpretation emerges without temporal supervision or handcrafted descriptors: neighborhoods are embedded independently from their geometry, yet the resulting clusters organize into spatially coherent motifs that evolve smoothly during crystallization. Based on this proof of concept, a refinement of the structural description of the liquid to account for its heterogeneity is planned in a subsequent work.

6 Conclusion

We presented a self-supervised method for clustering local atomic environments in 3D point clouds. By combining a rotation-equivariant encoder, invariant embeddings derived from equivariant features, and a VICReg objective on overlapping neighborhoods, the method learns symmetry-aware representations that remain directly clusterable. On a synthetic polycrystal benchmark, the learned embeddings outperform recent self-supervised point-cloud baselines as well as classical structural descriptors for unsupervised motif discovery. Applied to molecular dynamics simulations of aluminium crystallization, the method recovers spatially coherent clusters corresponding to liquid-like, ordered, and intermediate environments and reveals their evolution during nucleation and growth without labels or handcrafted order parameters. Finally, to assess generalization beyond the evaluation benchmark, we also perform a real-to-synthetic transfer experiment in which the encoder is pretrained on real MD trajectories and evaluated directly on the synthetic polycrystal dataset. Full details and results are provided in the supplementary material.

These results illustrate the potential of self-supervised geometric learning for large scientific point clouds, where local structure is subtle and difficult to capture with fixed descriptors. Future work will extend the approach to multi-species systems, incorporate temporal information during training, and further connect learned motifs with physical observables.

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